

Appendix C — Galois Connection Proof Sketch

This appendix gives the formal statement and proof sketch of the -approximate Galois connection between the category of Python program graphs and the category of GNN generative models. The informal version appears in Section 6.2 of the main text.

C.1 Categories

Let **Prog** be the category whose objects are typed Python program graphs $G = (V, E, _V, _E, _)$ in the sense of Section 2.2 (14 node kinds, 11 edge kinds) and whose morphisms are graph homomorphisms that preserve node and edge labels. Let **GNN** be the category whose objects are GNN v1.1 bundles (the Markdown sections `StateSpaceBlock`, `Connections`, `InitialParameterization`, `ActInfOntologyAnnotation`, plus the A/B/C/D matrices) and whose morphisms are role-preserving bundle embeddings.

Both categories are posets under the pointwise subset order: $G \leq G'$ in **Prog** iff $V \subseteq V'$, $E \subseteq E'$, and the labelings agree on the common subset; $M \leq M'$ in **GNN** iff each bundle section of M is included in the corresponding section of M' .

